

Multiple Choice (10 marks)

1. Statement 1 | For any two groups G and G' , there exists a homomorphism of G into G' . Statement 2 | Every homomorphism is a one-to-one map.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
2. If A is a subset of the real line $\mathbb{R}$ and A contains each rational number, which of the following must be true?
 - (a) If A is open, then $A = \mathbb{R}$.
 - (b) If A is closed, then $A = \mathbb{R}$.
 - (c) If A is uncountable, then $A = \mathbb{R}$.
 - (d) If A is uncountable, then A is open.
3. The element $(4, 2)$ of $\mathbb{Z}_{12} \times \mathbb{Z}_8$ has order
 - (a) 4
 - (b) 8
 - (c) 12
 - (d) 6
4. Suppose that $f(1 + x) = f(x)$ for all real x . If f is a polynomial and $f(5) = 11$, then $f(15/2)$
 - (a) -11
 - (b) 0
 - (c) 11
 - (d) $33/2$
5. For which of the following rings is it possible for the product of two nonzero elements to be zero?
 - (a) The ring of complex numbers
 - (b) The ring of integers modulo 11
 - (c) The ring of continuous real-valued functions on $[0, 1]$
 - (d) The ring $\{a + b \sqrt{2} : a \text{ and } b \text{ are rational numbers}\}$

True / False (10 marks)

Answer True or False

6. True or False: The order of the factor group $(\mathbb{Z}_{11} \times \mathbb{Z}_{15})/(\langle 1, 1 \rangle)$ is 5.
7. True or False: The element 3 is a generator for the finite field $\mathbb{Z}_{11}$.
8. True or False: The characteristic of the ring $\mathbb{Z}_3 \times \mathbb{Z}_3$ is 3, which is the smallest integer n such that $n * (1, 1) = (0, 0)$.
9. True or False: The intersection of two 2-dimensional subspaces of $\mathbb{R}^4$ must have a dimension of either 1 or 2.
10. True or False: The minimum value of the expression $x + 4z$, subject to the constraint $x^2 + y^2 + z^2 \leq 2$, is 0.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Divide the product of the first five positive composite integers by the product of the next five composite integers. Express your answer as a common fraction.
12. Discuss the implications of the composition of functions being injective. Given sets S , T , and U , and functions $f: S \rightarrow T$ and $g: T \rightarrow U$, analyze why $g \circ f$ being one-to-one necessitates that f is also one-to-one.

End of Paper